

3. Key Security Risks & Threats

A failure in endpoint security directly threatens the CIA Triad: Confidentiality, Integrity, and Availability. Our endpoint security design addresses and mitigates the following major risks:

- **Risk 1: Ransomware Propagation (The "Contractor Vector")**
 - **Scenario:** An unpatched contractor laptop connects to the company network to access the billing portal. Because the device lacks corporate antivirus, it introduces a wormable ransomware strain.
 - **Consequence:** The malware bypasses the perimeter and encrypts critical on-premise billing servers, halting operations and causing financial loss.
 - **Our Solution:** **Wazuh Agent** and **Host-Based Firewalls** isolate the infected device immediately upon detection, preventing spread.
- **Risk 2: Data Exfiltration (The "BYOD Gap")**
 - **Scenario:** A sales employee uses their personal unmanaged mobile phone to access corporate email. They inadvertently (or maliciously) copy a list of 500 VIP client phone numbers to their personal cloud storage or WhatsApp.
 - **Consequence:** Violation of Telecom privacy regulations (GDPR/local laws), leading to lawsuits and loss of reputation.
 - **Our Solution:** **Exchange ActiveSync Policies** prevent "Copy/Paste" between corporate email and personal apps.
- **Risk 3: Lateral Movement (The "Beachhead" Attack)**
 - **Scenario:** A standard employee laptop is compromised via a phishing link. The attacker uses this laptop not to steal data locally, but as a "jump box" to scan the internal network for vulnerable servers.
 - **Consequence:** The attacker gains access to the core infrastructure (Hybrid Cloud) using legitimate employee credentials.
 - **Our Solution:** **Host-Based Firewalls** (Block Inbound) and **Wazuh FIM** detect and block the scanning behavior.
- **Risk 4: Physical Theft (Data-at-Rest Exposure)**
 - **Scenario:** A field technician leaves their laptop in a service van, and it is stolen. The laptop contains cached network maps and customer addresses.
 - **Consequence:** If the hard drive is readable, the thief gains intelligence on the physical telecom infrastructure.
 - **Our Solution:** **BitLocker Encryption** (managed by Policy) ensures the data is mathematically unreadable without the TPM key.

- **Risk 5: Shadow IT & Unsanctioned Applications**
 - **Scenario:** To bypass file upload limits, an employee installs a free, unvetted "PDF Converter" or "File Sharer" found online. This software contains a "backdoor" Trojan or spyware.
 - **Consequence:** The software silently captures keystrokes or opens a port for attackers, completely bypassing traditional antivirus that relies on "known signatures."
 - **Our Solution: AppLocker Whitelisting** blocks *any* executable file that is not digitally signed by a trusted publisher (e.g., Microsoft or Adobe), preventing the installation entirely.
- **Risk 6: Privilege Escalation (The "Admin" Risk)**
 - **Scenario:** An attacker gains access to a developer's laptop. Because the developer has "Local Admin" rights to install software, the attacker uses those rights to disable the Antivirus and delete system logs to hide their tracks.
 - **Consequence:** Complete loss of visibility on that endpoint, allowing the attacker to dwell in the network undetected for months.
 - **Our Solution: Wazuh Security Configuration Assessment (SCA)** monitors the status of the Antivirus agent. If the AV is disabled, Wazuh triggers a "Critical Alert" to the SOC immediately.
- **Risk 7: Man-in-the-Middle (The "Coffee Shop" Attack)**
 - **Scenario:** A remote worker connects to the corporate cloud environment using unsecured public Wi-Fi at a cafe. An attacker on the same Wi-Fi network intercepts the session tokens.
 - **Consequence:** The attacker hijacks the active session to access cloud applications without needing the user's password.
 - **Our Solution: Host-Based Firewall** policies automatically block untrusted local network traffic when not connected to the secure corporate domain.
- **Risk 8: Zero-Day Vulnerabilities (The "Patch Gap")**
 - **Scenario:** A critical vulnerability is discovered in the Google Chrome browser (used by all 550 staff). It takes the IT team 3 days to manually test and approve the update.
 - **Consequence:** During those 3 days, any user visiting a malicious website could be compromised instantly.
 - **Our Solution: Tactical RMM** automates "Critical Severity" patches to bypass manual approval and install immediately, closing the window of opportunity for attackers.